

WEEK 8 – STUDENT PERFORMANCE DASHBOARD

Student Performance Analytics Dashboard using Microsoft Power BI

1. Introduction

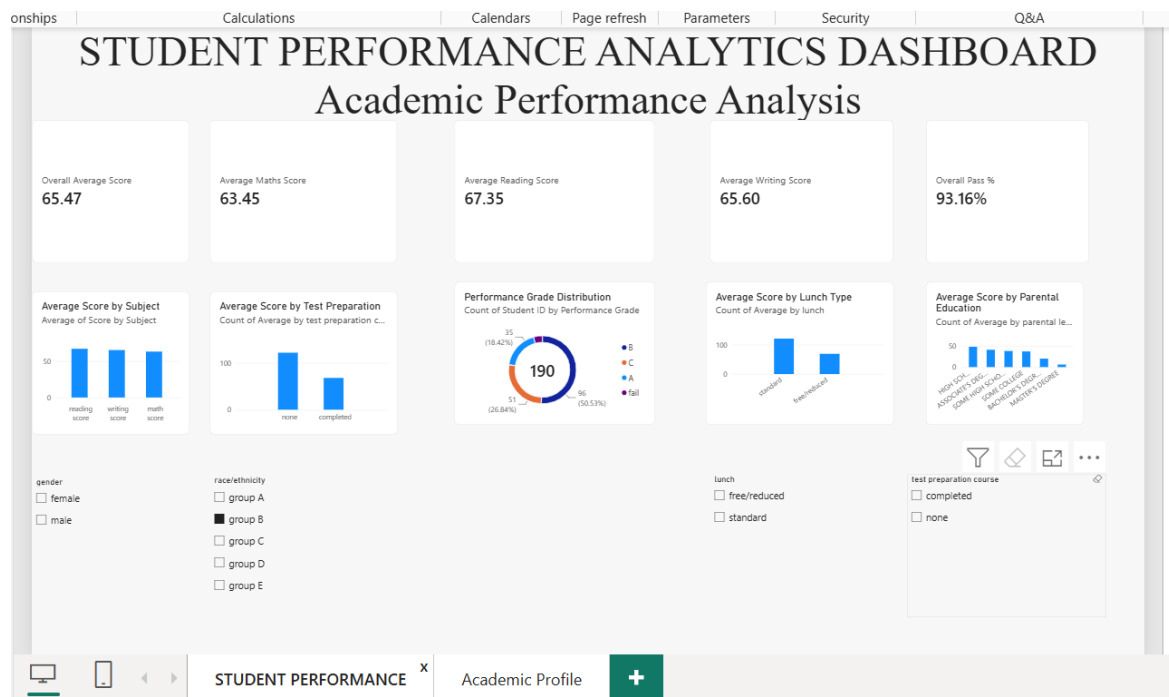
In Week 8, the prepared Student Performance dataset was used to build an interactive academic performance dashboard. The dashboard combines KPI cards, subject comparisons, demographic analysis, test-preparation analysis, lunch-type analysis, grade distribution, slicers and an Academic Profile drill-through page.

2. Dashboard Questions

The dashboard was designed to answer: What is the overall academic performance? Which subject has the highest or lowest average? Does test preparation relate to observed performance? How does parental education relate to scores? How does lunch type relate to performance? Which groups have higher or lower pass rates?

3. KPI Cards

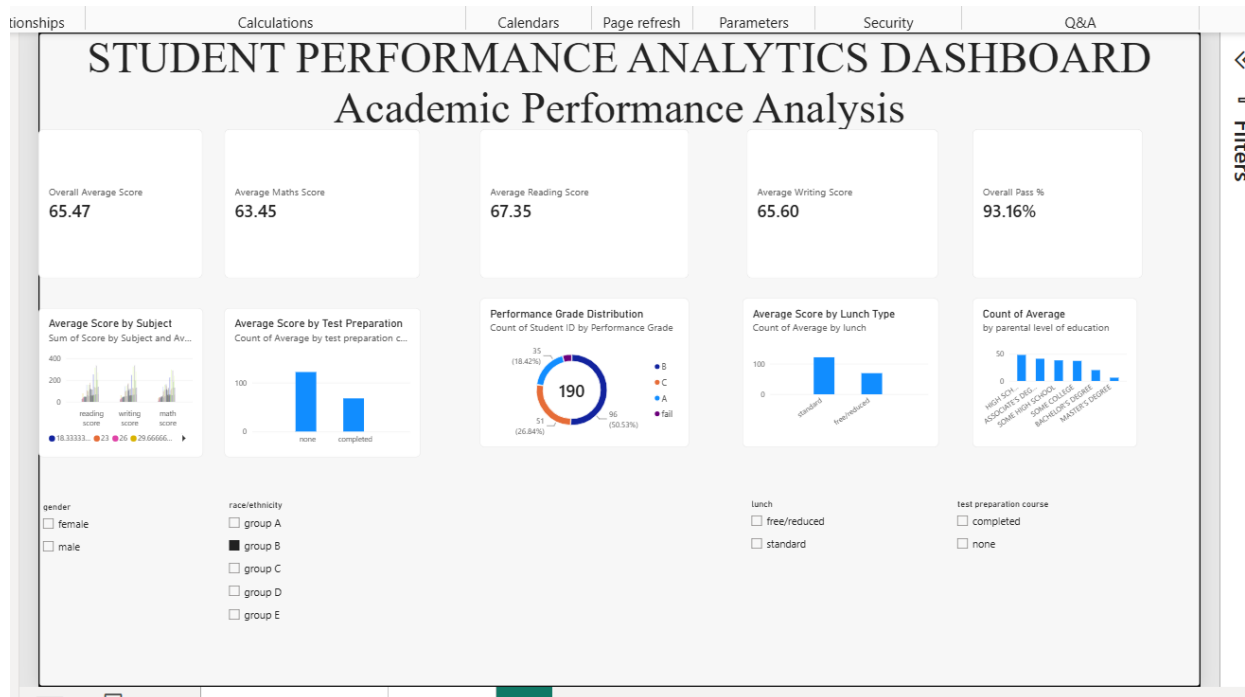
The dashboard includes KPI cards for Overall Average Score, Average Maths Score, Average Reading Score, Average Writing Score and Overall Pass %. These provide a quick summary of student performance.



Main Student Performance dashboard with KPI cards, charts and slicers.

4. Subject Performance

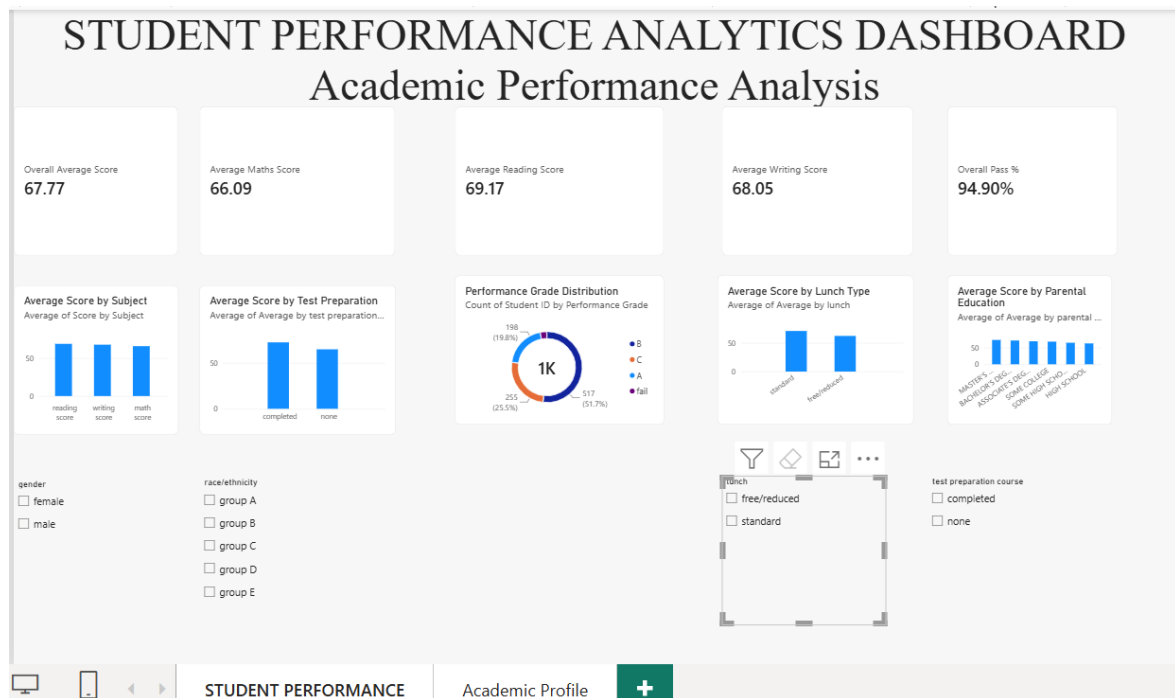
A clustered column chart compares the average scores of Reading, Writing and Math. The final chart uses one value series: Average Score, with Subject on the axis, so the three subjects can be compared directly.



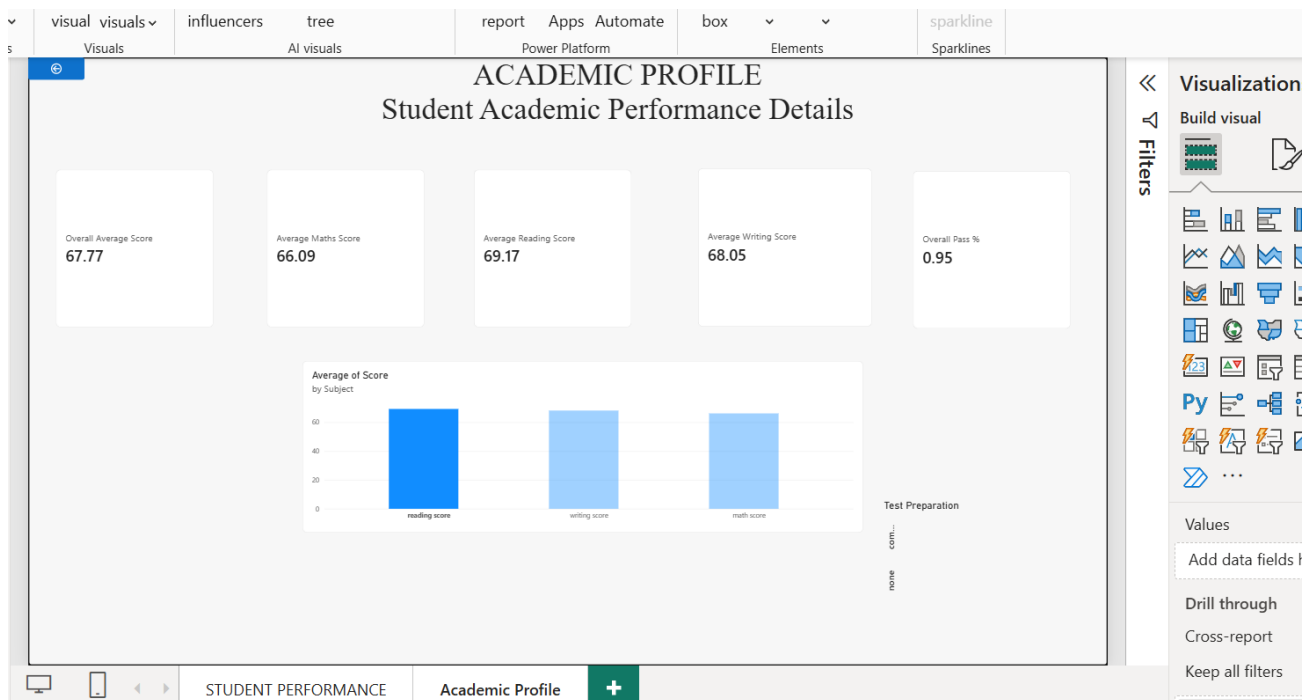
Dashboard layout showing the subject comparison and supporting visuals.

5. Supporting Analysis

The dashboard includes visuals for test preparation, performance grade distribution, lunch type and parental education. These visuals allow the user to compare observed average performance and student distributions across groups.



Dashboard view showing the completed subject, test-preparation, grade and lunch visuals.



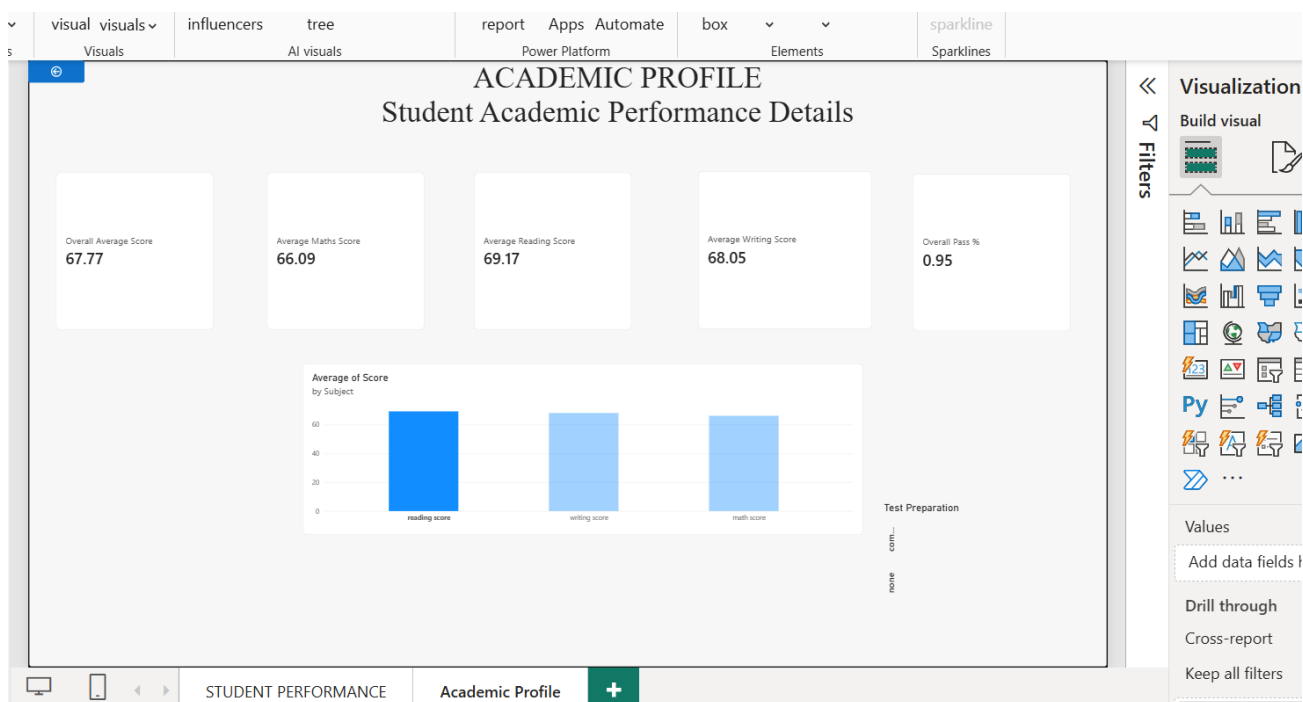
Academic Profile drill-through page with KPI cards and subject performance chart.

6. Interactive Slicers

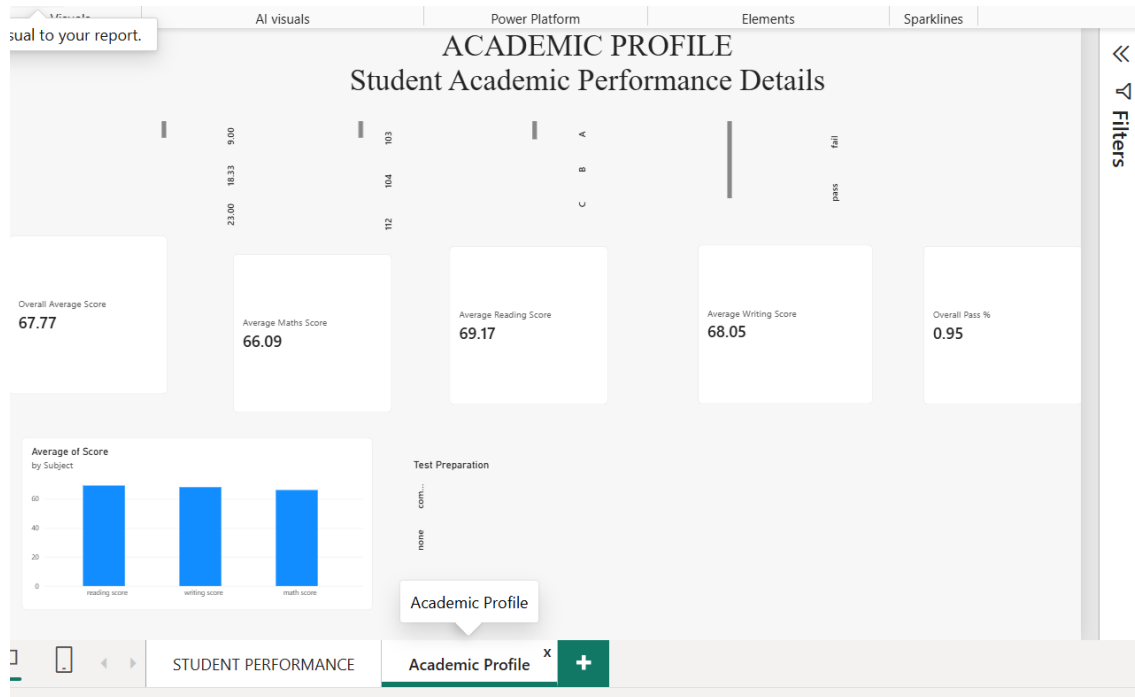
Slicers are provided for demographic and academic categories such as gender, race/ethnicity, lunch and test preparation course. Selecting a category filters the dashboard visuals for further exploration.

7. Academic Profile Drill-Through

A separate page named Academic Profile was created for detailed student academic performance. It is intended to show subject performance, average score, grade, pass/fail status and test-preparation information. A Back button can be used to return to the main dashboard.



Academic Profile drill-through page with KPI cards and subject performance chart.



Academic Profile page showing detailed student academic performance.

8. Evidence-Based Observations

- Reading has the highest average among the three displayed subject averages in the dashboard screenshot.
- Writing and Math are slightly lower than Reading in the shown dashboard.
- The dashboard shows different average performance for students who completed test preparation and those who did not.
- Lunch type groups show different observed average scores.
- Performance grades are distributed across A, B, C and fail categories.

9. Final Validation

The final dashboard contains KPI cards, subject comparison, test-preparation analysis, grade distribution, lunch-type analysis, parental-education analysis, slicers and the Academic Profile page. The final screenshots show the completed dashboard and drill-through layout.

10. Conclusion

Week 8 completed the Student Performance Analytics Dashboard. The prepared Week 7 data was converted into an interactive report that supports quick academic-performance comparison and further exploration through slicers and the Academic Profile page.